Reversible normalization and signed support

T-1185; 2026-10-31. Producer proof checks; independent source-fidelity review pending.

Polynomial normalization T_n(h)=a^(-n) h(aX+b) is reversible for a nonzero. The inverse uses parameters a^(-1), -b/a. Both compositions equal the original polynomial, and T_n is injective. Applying this inverse to the preceding validity transport proves an equivalence: Valid(T_6(f),T_deg(u)(u),T_3(v)) if and only if Valid(f,u,v). Here Valid means monic u, degree at most two, ordinate degree strictly below u, and u dividing f-v^2.

The signed affine map sends (x,y) to ((x-b)/a,y/a^3), with inverse (x,y) to (ax+b,a^3y). Both round trips are identities. Curve membership y^2=f(x) and signed support membership u(x)=0,y=v(x) are equivalent across the maps. Polynomial ordinate negation and point negation commute with normalization. These theorems hold over an arbitrary field with a nonzero; no prime-word bound is needed for this algebraic component.

Fourteen main declarations and eight controls are separately exposed for axiom readback. Rational fixtures use a=2,b=3: (5,8) maps to (1,1), (5,-8) to (1,-1), and the inverse recovers (5,8). Omitting y scaling fails the constant-polynomial curve condition; a=0 makes the point map noninjective. The constant-polynomial negative control is an algebraic equation fixture, not a smooth genus-two curve. Support u=X-5,v=8 is checked before and after transformation; a polynomial round trip is checked explicitly.

This establishes reversible affine coordinates and signed support predicates, not a Picard/Jacobian group isomorphism. Multiplicities, extension of the affine map at infinity, and principal-divisor transport remain the geometric construction to connect next. The source-level word/storage composition is in the preceding stored-divisor-validity packet. Existing compiled upstream proof dependencies remain in the trust boundary. This producer check does not supply independent review.

Sources: the prior Transport, Validity and Stored modules; canonical genus2_model_gate.rs with corrected named extraction; and Mathlib's polynomial algebra-map documentation (algEquivCMulXAddC and inverse substitution), Mathlib.Algebra.Polynomial.AlgebraMap (linked in REPORT.md). Elementary pinned polynomial composition identities were used without fetching compiled dependencies. No novelty claim.

All four final native checks passed with unchanged input hashes. Separate readback of all 22 declarations found only propext, Classical.choice and Quot.sound; status.json records the outcomes. The first Support attempt left a negated-division simplification unresolved; the explicit neg_div lemma fixes it. Receipts retain the failed attempt and nonfatal linter warnings. No scientific executions, timing observations or cost ratios.